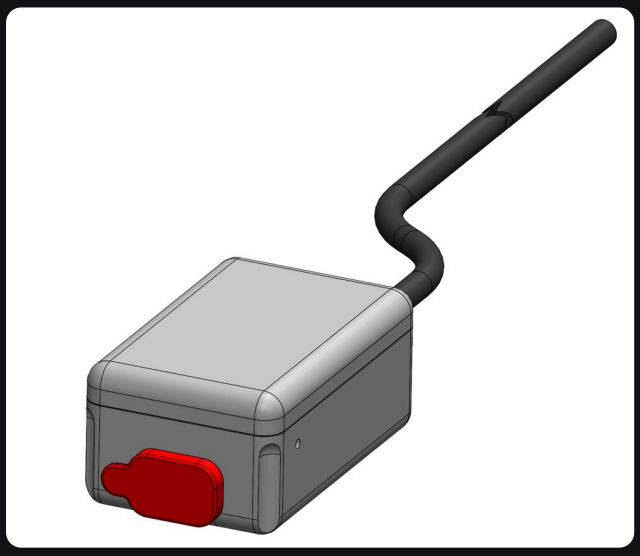
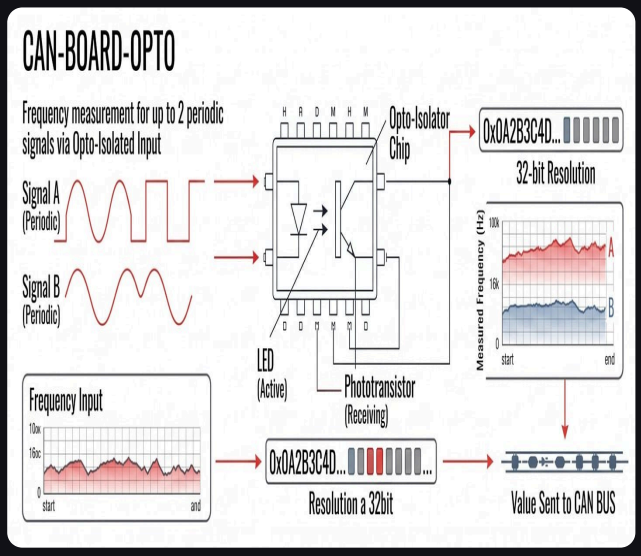
CAN-BOARD-OPTO

Technical datasheet and manual

2 optoisolated frequency inputs on CAN bus · Revision 0014 · November 2026





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1 Overview

1.1 Description

CAN-BOARD-OPTO measures the frequency of two periodic signals, through optoisolated inputs, and transmits it on the CAN bus with 32-bit resolution: for example engine speed, wheel speed, proximity sensors or flow meters.

CAN-BOARD boards are compact, programmable modules that acquire electrical or physical signals, from a sensor or an integrated module, process them and transmit them on the CAN bus. Each board is designed for a specific sensor or module; all of them share the same configuration parameters, set with the free **GUBELLINI DataStudio** software via USB-C. Once programmed, at every power-on the board measures and transmits the data according to its configuration.

1.2 Main features

- **2 optoisolated inputs**, frequency from **0 to 2 kHz**.
- **32-bit** resolution, 9 MHz counter.
- High-speed **CAN 2.0B** at 1 Mbps, 500 kbps or 250 kbps; programmable 11-bit or 29-bit IDs.
- Configuration via **USB-C** as a USB HID device, with no driver.
- Power supply from **7 to 24 V DC**, 80 mA.
- Integrated 120 Ω CAN termination, **AMP Super Seal 1.5** connector.
- Compact PA12 housing, **IPX7**, from **-20 °C to +85 °C**.

1.3 The CAN-BOARD family

CAN-BOARD boards are used in the laboratory, in machinery and plant monitoring, in precision agriculture, in process analysis, in automotive and in motorsport. Their data is easily recorded with the **GUBELOG** and **MICROLOG** data loggers as CAN channels (chap. 6).

Board	Measurement transmitted on the CAN
CAN-BOARD-ACCELEROMETER	3-axis accelerations, ±16 g
CAN-BOARD-GNSS	Latitude, longitude, ground speed, date and time
CAN-BOARD-LOADCELL	Signal of a load cell (Wheatstone bridge)
CAN-BOARD-OPTO	Frequency of 2 optoisolated pulse signals
CAN-BOARD-THERMOK	Temperature from a type K thermocouple
GUBENAV	High-dynamics GPS position and 6-axis IMU, up to 100 Hz

2 Technical data

2.1 Measurement

	Value
Measured quantity	Frequency of 2 periodic signals
Frequency range	From 0 to 2 kHz
Resolution	32 bits
Sampling	9 MHz
Format	Unsigned 32-bit integer

2.2 CAN bus

	Value
Transceiver	High-Speed CAN 2.0B
Bus speed	1 Mbps, 500 kbps, 250 kbps
ID format	11-bit (standard) or 29-bit (extended), freely chosen IDs
CAN IDs	Up to 4, 8 bytes each; uses ID 1
Transmission frequency	1, 10, 50 or 100 Hz
Data format	Intel (little endian)
Protocol	Open CAN (CANopen, EN 50325-4)
Termination	120 Ω resistor integrated in the module

> The transmission frequency is how many times per second the board sends its messages on the CAN; it does not change the sensor sampling rate, which is fixed (table in § 2.1). Avoid overloading the bus with needlessly high frequencies.

2.3 Power supply and USB-C

	Value
Supply voltage	From 7 to 24 V DC (pin 1 of the CAN connector)
Current consumption	80 mA
USB-C	On the back of the housing, protected by a rubber cap: sending and reading the configuration (USB HID, no driver). During configuration the board is powered by the USB.

2.4 Connectors

Power and CAN bus pass through the 4-way **AMP Super Seal 1.5 series** automotive connector (part number 282106-1), watertight.

CAN and power connector

Pin	Signal	Type
1	VBatt (7-24 V DC)	Power supply
2	CAN H	CAN bus
3	CAN L	CAN bus
4	GND	Ground

Sensor connector

Pin	Signal	Type
1	INPUT 1 +	Input 1 positive
2	INPUT 1 −	Input 1 negative
3	INPUT 2 +	Input 2 positive
4	INPUT 2 −	Input 2 negative

2.5 Operating environment

	Value
Degree of protection	IPX7
Operating temperature	From -20 °C to +85 °C

2.6 Dimensions and weight

	Value
Housing	Nylon PA12H
Length × width × height	66.2 × 38 × 27 mm (cables excluded)
Weight	80 g

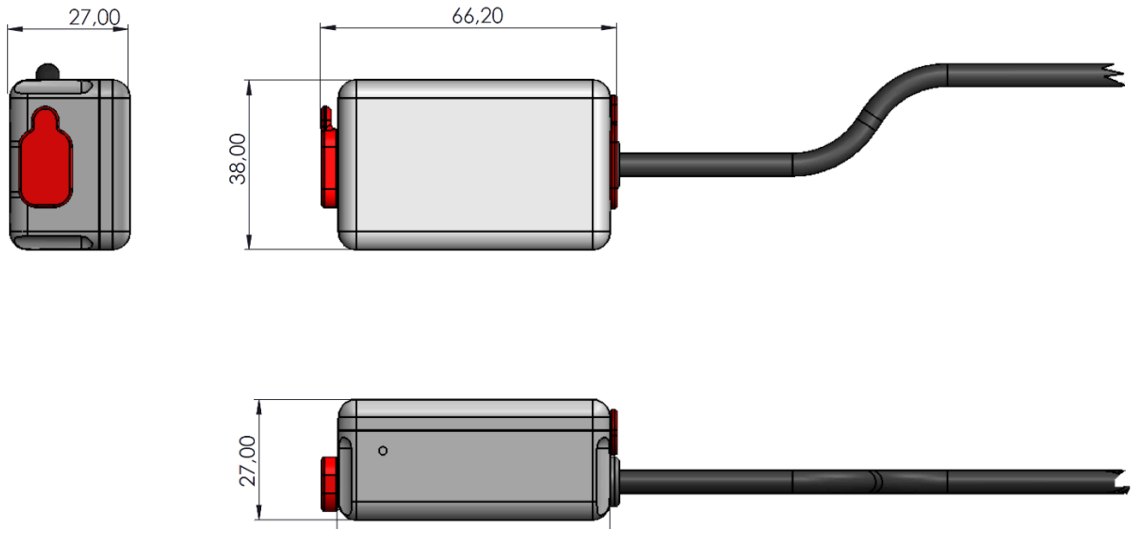


Figure 1 – CAN-BOARD dimensions (mm).

3 Connections

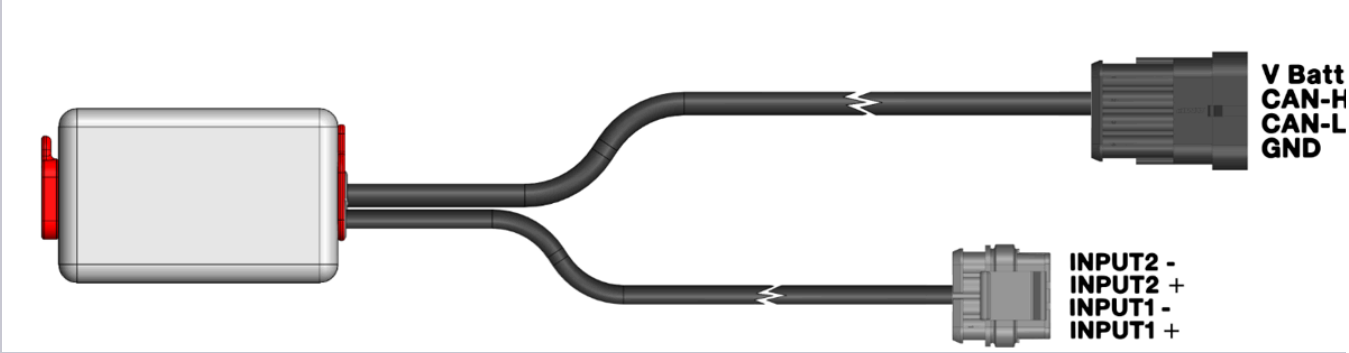


Figure 2 – CAN-BOARD-OPTO wiring diagram.

Connect the CAN connector to the CAN line of the data logger or system (CAN H to CAN H, CAN L to CAN L) and to the power supply. Several CAN-BOARD boards, GUBENAV and the data logger can share the same line: all of them must use the same speed and each board must transmit on different IDs.

Termination

The 120 Ω resistor is integrated in every CAN-BOARD. With a single board connected to a GUBELOG or a MICROLOG (also terminated) the line is correct; on larger networks check that the terminations are only at the ends of the line.

4 Configuration with GUBELLINI DataStudio

4.1 The software

Software	What it does
GUBELLINI DataStudio (free)	Configures Gubellini Electronics devices (CAN-BOARD, GUBENAV, GUBELOG, MICROLOG); imports signals from DBC files; downloads datasets from the data loggers and exports to MoTeC, AEM and CSV; includes an AI assistant for device connection and configuration.
GUBELLINI DataAnalysis (free)	Analyses the recorded datasets: plots versus time or distance, layouts with multiple plots, math channels, track map, on-board video; it includes an **AI assistant** that analyses the data, writes reports and generates the code of math channels.

4.2 Connection

1. Install DataStudio and connect the CAN-BOARD to the PC with the USB-C cable: the USB powers the board, which appears as a USB HID device (no driver is needed).
2. In the main window press **Sensor configuration** and choose **OPTO** in the list on the left.
3. **Read configuration** reads the parameters from the board; after making changes, **Send configuration** writes them to the flash memory, where they are kept even when the board is switched off. The log at the bottom reports the result of each operation.

> **One USB HID device at a time**
>
> The CAN-BOARD boards and MICROLOG appear to the PC in the same way: connect only one HID device at a time. **USB diagnostics** lists the devices seen by the PC.

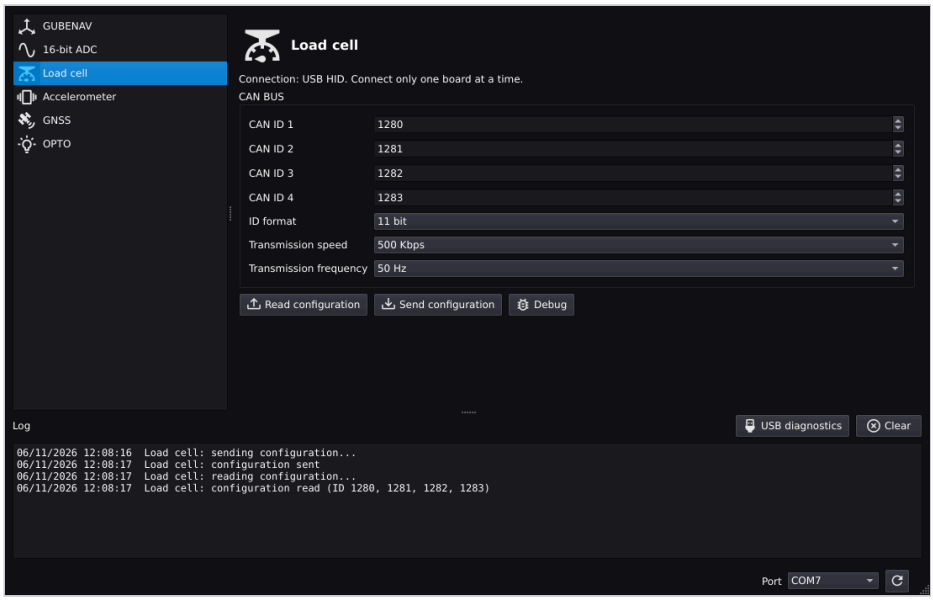


Figure 3 – A CAN-BOARD in Sensor configuration.

4.3 Parameters

Parameter	Meaning
CAN ID 1 - **CAN ID 4**	IDs on which the board transmits (this board uses ID 1). They are entered in **decimal**: for example 256 corresponds to 0x100. Choose IDs not used by other devices on the network.
ID format	11-bit (standard) or 29-bit (extended), according to the receiver.
Transmission speed	1 Mbps, 500 kbps or 250 kbps: the same for all devices on the bus.
Transmission frequency	1, 10, 50 or 100 Hz.

5 CAN protocol

5.1 Message content

Each message has 8 bytes; multi-byte values are in Intel (little endian) format, byte 0 is the least significant.

ID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
ID1	Frequency_1 bit 0-7	Frequency_1 bit 8-15	Frequency_1 bit 16-23	Frequency_1 bit 24-31	Frequency_2 bit 0-7	Frequency_2 bit 8-15	Frequency_2 bit 16-23	Frequency_2 bit 24-31
ID2	Not used	Not used	Not used	Not used	Not used	Not used	Not used	Not used
ID3	Not used	Not used	Not used	Not used	Not used	Not used	Not used	Not used
ID4	Not used	Not used	Not used	Not used	Not used	Not used	Not used	Not used

5.2 Conversion

Data	Position	Format	Conversion
Input 1 frequency Frequency_1	ID1, bits 0-31	Unsigned 32-bit	value / 100 = Hz
Input 2 frequency Frequency_2	ID1, bits 32-63	Unsigned 32-bit	value / 100 = Hz

6 Recording with GUBELOG and MICROLOG

The board data is recorded as **CUSTOM** CAN channels of a GUBELOG (**GUBELOG** button in the main window) or of a MICROLOG (**MICROLOG** button). In the logger set the same speed and the same ID format as the board, then set the channels as follows (Intel / little endian format, offset 0), with the IDs chosen in the board:

Channel	Label	CAN ID	Start bit	Bits	Type	Multiplier	Unit
CAN1	Frequency_1	256 (ID1)	0	32	Unsigned	0.01	Hz
CAN2	Frequency_2	256 (ID1)	32	32	Unsigned	0.01	Hz

Example: engine speed with 1 pulse per revolution

rpm = Hz × 60: in the logger channel use multiplier 0.01 × 60 = 0.6.

Frequencies

Set the logger channels to a frequency no higher than the board's transmission frequency: a higher frequency adds no information and takes up more space on the microSD. After making changes, send the configuration to the logger.

7 Troubleshooting

Problem	Cause and solution
No data on the CAN network	Check power supply and wiring (CAN H and CAN L not swapped), that the speed and ID format of the board are the same as the logger's and that the IDs in the logger channels are those set in the board.
The board does not appear in Sensor configuration	Use a USB-C data cable and connect only one USB HID device at a time; **USB diagnostics** lists the devices seen by the PC.
Stepped or duplicated values	The board's transmission frequency is lower than that of the logger channels: increase it or reduce the logger's.
Frequency always zero	Check the polarity of the optoisolated inputs (+ and −) and that the sensor signal is large enough to switch on the optoisolator.

8 Ordering information

Code	Description
GE-CB-0004	CAN-BOARD-OPTO – 2 optoisolated frequency inputs on CAN bus

Package contents: board, USB-C cable, instruction manual, GUBELLINI DataStudio and GUBELLINI DataAnalysis software.

8.1 Related products

Code	Description
GE-PROX-M8-3W-NO	M8 inductive proximity sensor (3-wire DC, N.O.)
GE-GL-0001 / 0003	GUBELOG-01 / GUBELOG-03 – 38-channel data logger with GPS, IMU, analog and digital inputs and CAN
GE-ML-0001	MICROLOG-01 – CAN logger with 16 CAN channels
GE-GN-0001	GUBENAV – 100 Hz high-dynamics GPS and IMU sensor, on CAN

Information and orders
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Technical data are subject to change without notice.